

# AgentDet: A Shared-Blackboard Multi-Agent Framework for Zero-/Few-Shot Object Detection

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## Abstract

*Large multimodal language models have made rapid progress on vision–language tasks, yet their potential for zero-/few-shot object detection (ZSOD/FSOD) under a closed set of target classes has yet to be fully explored. ZSOD/FSOD is hampered by data scarcity and catastrophic forgetting. Although vision–language models (VLMs) demonstrated excellent performance on multiple benchmarks, they typically rely on large-scale visual pre-training, which is inconsistent with the goal of FSOD to test generalization to new categories under limited supervision. We introduce AgentDet, a shared-blackboard multi-agent framework that unifies ZSOD and FSOD via pseudo-incremental learning. AgentDet decouples detection into four cooperating roles—Agent-Scout, Agent-Pinner, Agent-Curator, and Agent-Judge—which collaboratively maintain a Shared Blackboard and a Knowledge Base. For efficiency, we only train Agent-Judge by updating its image encoder and LLM-based detection head, which is a lightweight recipe that encourages generalization to previously unseen categories. On PASCAL VOC and MS COCO ZSOD/FSOD datasets, AgentDet achieves strongly competitive performance with state-of-the-art results in several settings.*

## 1. Introduction

ZS/FSOD are practical but challenging tasks that have attracted attention from researchers recently, aiming to detect objects without any, or with only a few, training images from novel classes. On the other hand, large Language Models (LLMs) [25, 39] are reshaping the patterns of multimodal research by leveraging their cross-modal comprehension and reasoning abilities. They have achieved remarkable performance on image-to-text generation [8] and visual question-resolution [21, 40] tasks. However, LLMs remain largely underexplored in ZS/FSOD tasks.

As shown in Fig. 1, existing FSOD methods rely on

finetuning or prototype learning. Traditional FSOD methods [4–6, 11, 16, 17, 19, 24, 27, 31, 33, 37] leverage metric learning, meta-learning, and data augmentation to improve model generalization. But such methods often remain susceptible to catastrophic forgetting and struggle to generalize robustly. On the other hand, visual-language models (VLMs) and recent LLM-based [9, 13, 18, 41] object detection methods or image segmentation methods offer promising zero-shot capabilities, which show promise but struggle to integrate external few-shot supervision. For example, HyperSeg [32] uses linguistic priors yet cannot effectively adapt to standard FSOD benchmarks, while FM-FSOD [13] relies on DINOv2 [26] without exploiting language reasoning ability of LLMs. VLMs like Grounding DINO achieve zero-shot performance but require large-scale pretraining incompatible with standard ZS/FSOD protocols (Fig. 1(b)). Moreover, most prior works treat zero-shot and few-shot tasks separately, lacking a unified framework.

We draw inspiration from cinema: when no photo of a target person exists (as in the montage scene of *The Truman Show*), one can cut eyes, nose, and mouth from magazine headshots and assemble a face that resembles the target person (or the way police construct a suspect sketch from witness-described features). In analogy, an LLM often behaves like a “blind expert”, a capable of rich verbal descriptions without truly seeing. This motivates a recipe tailored to ZS/FSOD: accumulate fragment-level visual evidence, compose it into plausible holistic textual cues, and make bounding-box decisions guided by LLM’s language prior knowledge. Because novel classes in real applications are frequently unlabeled, the process is pseudo-incremental: the system writes high-consistency local evidence into a knowledge base without new-class box annotations and later aligns with a few labels, without changing the architecture.

Building on this insight, we present AgentDet, a shared-blackboard multi-agent framework that unifies ZS/FSOD via pseudo-incremental knowledge base maintaining. The system revolves around a Shared Blackboard and a Knowledge Base(KB). Agent-Scout ingests the image

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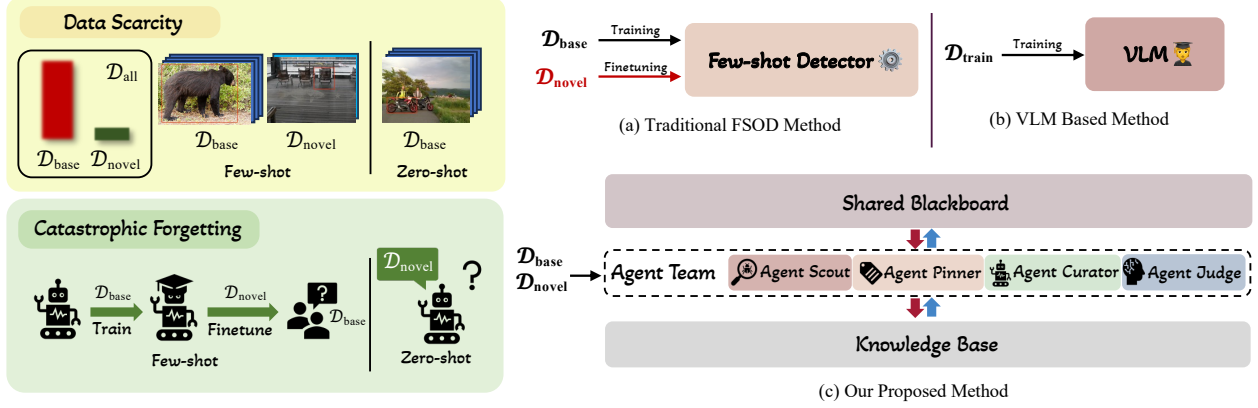


Figure 1. ZS/FSOD challenges and the traditional method. **Left:** Two challenges—data scarcity and catastrophic forgetting. **Right:** Representative methods: (a) Traditional transfer learning pipelines requiring two-stage training and finetuning; (b) zero-shot detectors based on VLMs, which rely on large-scale visual pre-training but adapt poorly on the FSOD benchmarks; (c) AgentDet, a multi-agent system with a shared blackboard that aggregates evidence regarding both base and new classes at the fragment level into a unified knowledge base.

and produces holistic textual cues; Agent-Pinner retrieves fragment-level visual references from the KB and pins them onto the Shared Blackboard; Agent-Curator maintains and updates the KB according to the information on the Shared Blackboard; Agent-Judge reads the Shared Blackboard then outputs the final bounding boxes and confidences. The Shared Board is the state space jointly maintained by the four agents, while Agent-Pinner and Agent-Curator interact bidirectionally with the KB. In training stage, Training uses a single unified detection loss and updates only Agent-Judge’s image encoder and LLM-based detection head which adapted with LoRA, which is a lightweight pipeline that improves generalization to previously unseen categories. Confronting data scarcity, we fragment visual evidence and learn it in a unified space to endow the model with reasoning ability on unseen categories; confronting catastrophic forgetting, the pseudo-incremental training regime unifies ZSOD and FSOD within one architecture and execution loop.

During training, ground-truth object annotations are available. We first crop the annotated regions and then apply over-segmentation method within each crop to obtain high-purity patches. At inference time, no object annotations are provided for novel classes so we directly over-segment the entire query image and rely on retrieval and consistency gating to select informative fragments. Because the system updates knowledge from unlabeled images—without new-class bbox supervision, we refer to this process as pseudo-incremental learning. Our contributions are threefold:

- **AgentDet Framework.** We propose a shared-blackboard multi-agent framework tailored to ZS/FSOD, where four specialized agents collaborate via a Shared Board and a KB to provide holistic textual cues, fragment-level references, safe KB updates, and final detections.

- **Unification of zero-/few-shot detection.** Through a pseudo-incremental learning schedule, our approach unifies ZS/FSOD within a single architecture and execution loop, enabling a smooth transition from zero to few shots without architectural changes.
- **Comprehensive evaluation.** Extensive experiments on multiple benchmark datasets demonstrate that AgentDet achieves superior performance, proving its efficiency and robustness in zero/few-shot detection scenarios.

## 2. Related Works

Recent progress in large language models (LLMs) has led to strong performance on tasks involving semantic reasoning and retrieval. Proprietary systems such as GPT-4 [1] and ERNIE Bot [3] have shown this potential in real-world use. However, because these models are not fully open, it is difficult for academic researchers to study them in depth or reproduce their behavior. In contrast, open-source LLMs such as Llama [10, 28], Qwen [2, 14], and DeepSeek [7] provide a more practical basis for research because their weights and technical details are publicly available. In our experiments, in order to leverage the rich prior knowledge of LLMs, we use Llama3.1-8B-Instruct, Qwen2.5-7B, and Qwen3-8B as the LLM backbones of AgentDet.

**Multimodal Large Language Model.** Recent multimodal large language models (MLLMs) have made cross-modal understanding much stronger by combining visual and language information, and in some cases audio as well. General-purpose systems such as LLaVA [21, 22], GPT-4o [15], and Qwen-VL [30] show strong ability on open-domain visual reasoning tasks. Meanwhile, there are also domain-oriented models. For example, Valley [35], Med-PaLM [29], and Open-FinLLMs [36], incorporate task-

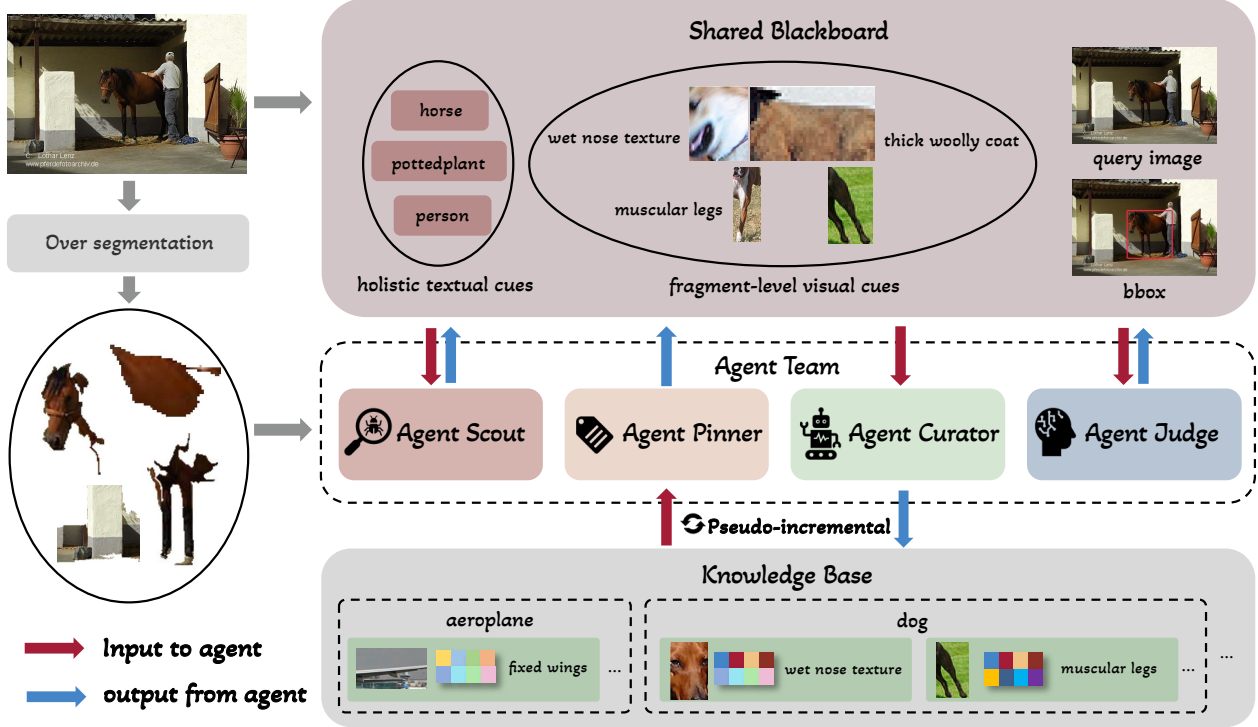


Figure 2. AgentDet overview. The query image is processed by four agents collaborating: *Scout* writes holistic textual cues ( $H$ ), *Pinner* pins fragment-level visual cues retrieved from the Knowledge Base ( $R$ ), *Curator* performs pseudo-incremental safe updates to the KB, and *Judge* reads the board to output final detections ( $B, \sigma$ ).

specific knowledge for particular applications. Still, these models are not designed for general object detection settings. More importantly, most existing MLLMs focus on image-level understanding rather than explicit instance-level localization, such as object detection or segmentation. As a result, their use in ZS/FSOD remains relatively limited.

**Zero/Few-shot Object Detection.** Conventional FSOD methods mainly build on transfer learning [24, 31], meta-learning [23], and data augmentation [11]. Although these strategies have been widely studied, they often struggle when only a few novel-class samples are available, especially when the gap between base and novel classes is large. Training can also be unstable in episodic settings. More recent methods built on foundation models still have clear limitations. LLaFS relies on a fixed CodeLlama backbone and requires polygon-level supervision, which makes it unsuitable for zero-shot settings. FM-FSOD [13], on the other hand, mainly uses LLMs as static classifiers and does not make full use of their reasoning or language capabilities.

**LLM-era Agents and Multi-Agent Systems.** Recent work on LLM-based agents [12] has roughly followed three lines. The first studies reason-act loops, where

chain-of-thought(CoT) is interleaved with tool use, as in ReAct-style agents [38]. The second focuses on self-reflection and memory, using natural-language feedback to improve behavior across multiple rounds. The third explores multi-agent role division, where specialized agents interact through dialogue or programmatic coordination, as in AutoGen/CAMEL-style systems [20, 34]. From the perspective of training, many such methods follow centralized training and decentralized execution (CTDE), which allows coordination to be learned jointly while keeping inference modular.

### 3. Method

#### 3.1. Problem Definition

In ZS/FSOD tasks, the set of object categories  $\mathcal{C}$  is divided into base classes  $\mathcal{C}_b$  and novel classes  $\mathcal{C}_n$ , such that:  $\mathcal{C} = \mathcal{C}_b \cup \mathcal{C}_n$ ,  $\mathcal{C}_b \cap \mathcal{C}_n = \emptyset$ . The base dataset  $\mathcal{D}_{\text{base}}$  contains a large number of training images from base classes  $\mathcal{C}_b$ :  $\mathcal{D}_{\text{base}} = \{A_1^b, A_2^b, \dots, A_{|\mathcal{C}_b|}^b\}$ . The novel dataset  $\mathcal{D}_{\text{novel}}$  contains only a limited number of training images from novel classes  $\mathcal{C}_n$ , where each class  $i$  has exactly  $K$  instances:

$$\mathcal{D}_{\text{novel}} = \{A_1^n, A_2^n, \dots, A_{|\mathcal{C}_n|}^n\}, |A_i^n| = K, \forall i \in [1, |\mathcal{C}_n|]. \quad (1)$$

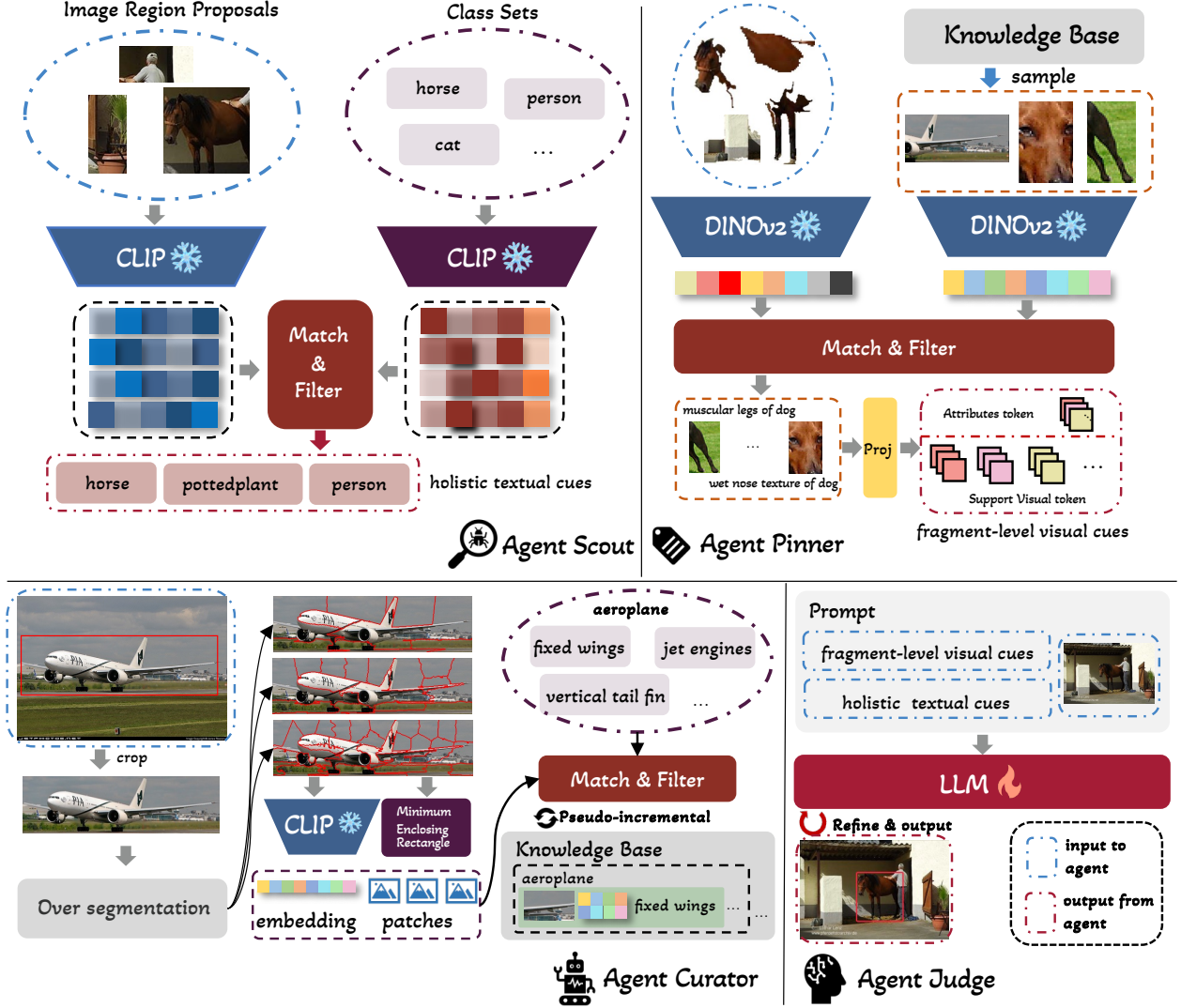


Figure 3. AgentDet pipeline details. **Top-left (Agent-Scout):** CLIP matches region proposals to class prompts, writing holistic textual cues  $H$ . **Top-right (Agent-Pinner):** DINOv2 retrieves KB patches and composes fragment-level cues  $R$ . **Bottom-left (Agent-Curator):** pseudo-incremental crop-oversegmentation with Match&Filter maintains the KB. **Bottom-right (Agent-Judge):** an LLM consumes  $(R, H)$  with visual tokens and outputs final detections  $(B, \sigma)$ .

Here,  $K = 0$  corresponds to ZSOD, where the model detects novel objects without any training images. When  $K > 0$ , the task is referred to as FSOD, where only a limited number of labeled images are available for each novel class.

### 3.2. Shared Blackboard and Knowledge Base

We distinguish two complementary memory components: a persistent *Knowledge Base* (KB) for long-term visual evidence and a transient *Shared Blackboard* (Board) for per-image coordination. The KB, denoted by  $\mathcal{K}$ , stores patch-level entries enriched with visual embeddings, spatial descriptors, and semantic attributes; it supports patch-level retrieval. The Board, denoted by  $\mathcal{M}_{\text{board}}$ , is a low-latency

state space that aggregates cues and control flags posted by agents during a single inference episode.

At time step  $t$ , we represent the Board state as

$$\mathcal{M}_{\text{board}}^{(t)} = \left\{ H^{(t)}, R^{(t)}, B^{(t)}, \Sigma^{(t)}, r_{\text{scout}}^{(t)}, r_{\text{pinner}}^{(t)}, \text{pm}^{(t)} \right\} \quad (2)$$

where  $H$  are *holistic textual cues* (the target set  $\mathcal{C}_{\text{target}}$  with scores),  $R$  are *fragment-level references* retrieved from  $\mathcal{K}$  (e.g.,  $[\phi_k^{\text{rect}} \oplus \phi_k^{\text{attr}}]$ ),  $(B, \Sigma)$  are the predicted boxes and confidences,  $r_{\text{scout}}, r_{\text{pinner}} \in \{0, 1\}$  indicate agent readiness,  $\text{pm} \in \{0, 1\}$  gates safe KB updates.

Let  $\mathcal{C}_{\text{target}}$  be produced from CLIP-based alignment and



posterior verification, and let  $\mathcal{R}$  be the index set of retrieved patches from  $\mathcal{K}$  (see Sec. 3.5). We define a simple consistency gate:

$$\text{cons}(k, c) = \mathbf{1} \left[ \max_i \mathbf{M}_{i,k} > \alpha_{\text{search}} \right] \cdot \mathbf{1} \left[ \max_i \mathbf{S}_{i,c} > \tau \right], \quad (3)$$

where  $\mathbf{M}$  is the query–KB similarity matrix and  $\mathbf{S}$  is the crop–class similarity matrix. A safe write is permitted when there exists  $(k, c) \in \mathcal{R} \times \mathcal{C}_{\text{target}}$  with  $\text{cons}(k, c) = 1$  and the attribute score  $s_{p,a} > \alpha_{\text{seg}}$ :

$$\text{pm} = 1 \iff \exists (k, c) : \text{cons}(k, c) = 1 \wedge s_{p,a} > \alpha_{\text{seg}}. \quad (4)$$

Otherwise  $\text{pm} = 0$ , and entries are deferred or decayed.

### 3.3. Agent-Scout

Within our shared-blackboard multi-agent system, *Agent-Scout* serves as the generator of *holistic textual cues* for a query image  $I \in \mathbb{R}^{H \times W \times 3}$ . It performs dense region cropping followed by CLIP-based visual–semantic alignment to obtain a prioritized set  $\mathcal{C}_{\text{target}} \subseteq \mathcal{C}$  that guides downstream patch-level retrieval and grounding. The resulting cues are posted to the *Shared Blackboard* for other agents to consume.

We first construct a set of region proposals  $\mathcal{B}$  by applying uniform cropping over multiple spatial grids and scales:

$$\mathcal{B} = \bigcup_{\substack{r \in \mathcal{R} \\ s \in \mathcal{S}}} \bigcup_{m=1}^M \mathcal{T}_r(I; \alpha_s), \quad (5)$$

where  $\mathcal{R}$  denotes the set of aspect ratios (e.g.,  $(1, 1), (1, 2), (2, 1)$ ), and  $\mathcal{S}$  denotes the sampling ranges of scale factors. Each transformation  $\mathcal{T}_r(\cdot; \alpha_s)$  extracts  $M$  regional patches under ratio  $r$  and scale  $\alpha_s$ .

We encode each patch  $I_b$  using CLIP’s visual encoder, and each class name  $c_j \in \mathcal{C}$  with the textual prompt “a photo of a  $c_j$ ” using CLIP’s text encoder:

$$\Phi_{\text{vis}} = f_{\text{CLIP}}^{\text{img}}(\{I_b\}_{b \in \mathcal{B}}), \quad (6)$$

$$\Phi_{\text{txt}} = f_{\text{CLIP}}^{\text{txt}}(\{"a photo of a " \oplus c_j\}_{j=1}^N). \quad (7)$$

We compute a similarity matrix  $\mathbf{S} \in \mathbb{R}^{|\mathcal{B}| \times N}$  between cropped patches and class prompts, scaled by a temperature coefficient  $\kappa = 100$ :

$$\mathbf{S} = \kappa \cdot \text{softmax}(\Phi_{\text{vis}} \Phi_{\text{txt}}^\top). \quad (8)$$

We then define the set of detected categories as those with high similarity across any region:

$$\mathcal{C}_{\text{detected}} = \left\{ c_j \mid \max_i \mathbf{S}_{i,j} > \tau \right\}, \quad (9)$$

and apply an optional posterior verification step using the LLM to produce the final target category set:

$$\mathcal{C}_{\text{target}} = \{c \in \mathcal{C}_{\text{detected}} \mid P(c|I) > \gamma\}. \quad (10)$$

In practice, we treat  $\mathcal{C}_{\text{target}}$  (together with  $\{\max_i \mathbf{S}_{i,j}\}_j$  and  $P(c|I)$ ) as the *holistic textual cues* written by Agent-Scout onto the Shared Blackboard, which then provide semantic guidance for the subsequent patch-level retrieval module (Sec. 3.5).

### 3.4. Agent-Curator

*Agent-Curator* constructs and maintains a structured visual Knowledge Base  $\mathcal{K}$  composed of patch-level entries enriched with semantic attributes and visual features. Each entry encodes localized visual knowledge grounded on open-vocabulary concepts. Given a labeled dataset  $\mathcal{D} = \{(I_n, \mathcal{A}_n)\}_{n=1}^N$ , where  $\mathcal{A}_n$  denotes the set of annotated categories for image  $I_n$ , our goal is to build:

$$\mathcal{K} = \left\{ (p_k, \phi_k^{\text{attr}}, \phi_k^{\text{rect}}, s_k) \right\}_{k=1}^K, \quad (11)$$

where each entry stores the visual patch  $p_k$ , its attribute embedding  $\phi_k^{\text{attr}}$ , bounding box coordinate  $\phi_k^{\text{rect}}$ , and a CLIP-based matching score  $s_k$ .

We extract dense patches from each image via multi-scale oversegmentation:

$$\mathcal{P}_n = \bigcup_{s \in \mathcal{S}} \mathcal{U}(I_n; \lambda_s), \quad (12)$$

where  $\mathcal{S} = \{\text{sml}, \text{med}, \text{lg}\}$  denotes scale presets and  $\lambda_s$  controls segment granularity. For each patch  $p \in \mathcal{P}_n$ , we extract its visual embedding and bounding box:

$$\phi_p^{\text{vis}} = f_{\text{CLIP}}(p), \quad \phi_p^{\text{rect}} = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]. \quad (13)$$

To enrich each patch with semantic attributes, we generate a list of textual attributes  $\mathcal{A}_c = \{a_{c,m}\}_{m=1}^{M_c}$  for each class  $c \in \mathcal{C}$  and match every attribute  $a$  to patch  $p$  via CLIP cosine similarity:

$$s_{p,a} = \frac{\langle f_{\text{CLIP}}^{\text{txt}}(a), \phi_p^{\text{vis}} \rangle}{\|f_{\text{CLIP}}^{\text{txt}}(a)\| \cdot \|\phi_p^{\text{vis}}\|}. \quad (14)$$

For each class  $c$ , we store the top-3 matched attributes with score above a threshold  $\alpha_{\text{seg}}$ :

$$\mathcal{K}_c = \left\{ (\phi_p^{\text{rect}}, a, s_{p,a}) \mid s_{p,a} > \alpha_{\text{seg}} \right\}_{\text{top-3}}. \quad (15)$$

To control memory and maintain semantic diversity, we retain at most 50 entries per class based on the scores  $s_{p,a}$  using a “last-place elimination” policy:

$$\mathcal{K} \leftarrow \bigcup_c \text{Top}_{|\mathcal{K}_c| \leq 50}(\mathcal{K}_c). \quad (16)$$

The resulting  $\mathcal{K}$  is class-agnostic, interpretable, and supports patch-level retrieval.

**Pseudo-incremental.** When ground-truth boxes are available during training, we first crop each annotated region and then run over-segmentation  $\mathcal{U}$  inside the crop to obtain higher-purity patches. Let  $\mathcal{B}_n$  denote the set of ground-truth boxes for  $I_n$  and  $\text{crop}(I_n, b)$  be the image restricted to box  $b$ . We form:

$$\mathcal{P}_n^{\text{train}} = \bigcup_{b \in \mathcal{B}_n} \bigcup_{s \in \mathcal{S}} \mathcal{U}(\text{crop}(I_n, b); \lambda_s), \quad (17)$$

which reduces attribute noise and background contamination in the entries committed to  $\mathcal{K}$ . At inference time, no object annotations are provided for novel classes; we therefore over-segment the *entire* image:

$$\mathcal{P}_q^{\text{test}} = \mathcal{U}(I_q; \{\lambda_s\}), \quad (18)$$

retrieve fragments against  $\mathcal{K}$ , and rely on the consistency gate (Eq. 3) together with the attribute agreement and the safe-write rule (Eq. 4) to decide whether a candidate update is committed, deferred, or decayed (with capacity control via last-place elimination). This asymmetry—*crop-then-oversegment* with labels at training versus *full-image over-segmentation* without labels at inference—allows the Curator to continually densify the KB from unlabeled streams *without* new-class box supervision. We refer to this regime as *pseudo-incremental* maintenance: the KB improves over time from consistent evidence, and once a few labeled instances arrive, the same pipeline seamlessly supports FSOD without architectural changes.

### 3.5. Agent-Pinner

Within our shared-blackboard system, *Agent-Pinner* retrieves *fragment-level visual cues* from the Knowledge Base  $\mathcal{K}$  and pins them onto the Shared Blackboard as references for grounding and detection. Given a query image  $I_q$  and the constructed  $\mathcal{K}$ , *Agent-Pinner* returns a reference representation  $\Phi_{\text{ref}}$  to be fed into the generation module:

$$I_q, \mathcal{K} \longrightarrow \Phi_{\text{ref}}. \quad (19)$$

We first segment the input image using the same over-segmentation operator  $\mathcal{U}$  as in KB construction:

$$\mathcal{P}_q = \mathcal{U}(I_q; \{\lambda_s\}), \quad \mathbf{Q} = f_{\text{DINOv2}}(\mathcal{P}_q), \quad (20)$$

where  $f_{\text{DINOv2}}$  encodes each query patch into a dense visual embedding. We compute pairwise similarities between each query patch and all entries in  $\mathcal{K}$  using dot product:

$$\mathbf{K} = f_{\text{DINOv2}}(\mathcal{K}), \quad \mathbf{M} = \mathbf{Q}\mathbf{K}^\top. \quad (21)$$

To filter out irrelevant background regions, we select a subset of retrieved patches whose similarity exceeds a threshold:

$$\mathcal{R} = \left\{ k \mid \max_i \mathbf{M}_{i,k} > \alpha_{\text{search}} \right\}. \quad (22)$$

Finally, the retrieved patches  $\{k\}$  are converted into a reference sequence by concatenating their spatial and semantic descriptors:

$$\Phi_{\text{ref}} = g_\theta \left( \left[ \phi_k^{\text{rect}} \oplus \phi_k^{\text{attr}} \right]_{k \in \mathcal{R}} \right), \quad (23)$$

where  $g_\theta(\cdot)$  is a learnable projection (e.g., MLP/adaptor) and  $\oplus$  denotes vector concatenation. In execution, *Agent-Pinner* writes  $\Phi_{\text{ref}}$  and the indices  $\mathcal{R}$  onto the Shared Blackboard as pinned references  $R$ , which are consumed by the Decision agent for final detection.

### 3.6. Agent-Judge

As the decision-making agent in our shared-blackboard system, *Agent-Judge* consumes the holistic textual cues  $\mathcal{C}_{\text{target}}$  (from *Agent-Scout*) and the fragment-level references  $\Phi_{\text{ref}}$  (from *Agent-Pinner*) posted on the Shared Blackboard, and produces the final detections. After obtaining  $\Phi_{\text{ref}}$  and  $\mathcal{C}_{\text{target}}$ , *Agent-Judge* performs language-based inference followed by a lightweight self-refinement loop to improve localization quality and reliability. The procedure is detailed below while preserving the original computation.

Given the query image  $I_q$ , category candidates  $\mathcal{C}_{\text{target}}$ , and reference input  $\Phi_{\text{ref}}$ , the model generates initial bounding box predictions  $\mathcal{B}_{\text{init}} = \{b_i\}_{i=1}^K$ . We first encode the query image using an EVA-CLIP encoder followed by a Q-Former:

$$\mathbf{V} = f_{\text{QFormer}}(f_{\text{EVA-CLIP}}(I_q)). \quad (24)$$

We then assemble a language prompt with explicit task description, the retrieved reference knowledge, and output specification. The prompt  $\mathcal{P}$  is passed to the LLM (parameterized by  $\theta_{\text{LoRA}}$ ) together with the visual tokens  $\mathbf{V}$ :

$$\mathcal{B}_{\text{init}} = f_{\text{LLM}}^{\theta_{\text{LoRA}}}(\mathcal{P}; \mathbf{V}). \quad (25)$$

To further enhance localization accuracy, we apply a self-refinement step based on prompting. A new prompt  $\mathcal{P}_{\text{refine}}$  is constructed by incorporating the current detections, and the LLM produces refined predictions:

$$\mathcal{B}_{\text{final}} = f_{\text{LLM}}^{\theta_{\text{LoRA}}}(\mathcal{P}_{\text{refine}}; \mathbf{V}). \quad (26)$$

The confidence  $\sigma_i$  of each predicted object is computed as the sequence probability over token outputs:

$$\sigma_i = \prod_{t=1}^T P(w_t^{(i)} \mid w_{1:t-1}^{(i)}; \theta). \quad (27)$$

We filter predictions with a confidence threshold  $\beta$ :

$$\mathcal{B}_{\text{final}} = \{(b_i, \sigma_i) \mid \sigma_i > \beta\}_{i=1}^{K'}. \quad (28)$$

In execution, *Agent-Judge* writes  $(\mathcal{B}_{\text{final}}, \{\sigma_i\})$  back to the Shared Blackboard (optionally with conflict flags), enabling *Agent-Curator* to perform safe updates to the Knowledge Base under the pseudo-incremental schedule.

| Method                            |                  | Novel Split 1 |             |             |             | Novel Split 2 |             |             |             | Novel Split 3 |             |             |             |
|-----------------------------------|------------------|---------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|
|                                   |                  | 1             | 3           | 5           | 10          | 1             | 3           | 5           | 10          | 1             | 3           | 5           | 10          |
| FSRW                              | <i>ICCV19</i>    | 14.8          | 26.7        | 33.9        | 47.2        | 15.7          | 22.7        | 30.1        | 40.5        | 21.3          | 28.4        | 42.8        | 25.9        |
| TFA                               | <i>ICML20</i>    | 36.8          | 43.6        | 55.6        | 56.1        | 23.5          | 34.1        | 35.1        | 39.1        | 20.6          | 29.4        | 43.9        | 44.1        |
| FSCE                              | <i>CVPR21</i>    | 44.2          | 51.4        | 57.3        | 58.9        | 27.3          | 43.5        | 44.4        | 50.2        | 37.2          | 47.5        | 54.6        | 58.5        |
| FADI                              | <i>NeurIPS21</i> | 50.3          | 56.5        | 59.1        | 63.4        | 30.6          | 40.3        | 42.8        | 47.9        | 45.7          | 49.1        | 55.0        | 59.6        |
| LVC                               | <i>CVPR22</i>    | 53.4          | 58.3        | 59.1        | 62.9        | 32.8          | <u>50.7</u> | 49.3        | 50.1        | 48.4          | 54.9        | 59.6        | 59.6        |
| MRSN                              | <i>ECCV22</i>    | 46.1          | 57.9        | 63.1        | 65.1        | 31.2          | 46.7        | 47.1        | 50.6        | 35.5          | 45.6        | 54.4        | 57.4        |
| ICPE                              | <i>AAAI23</i>    | 54.1          | 62.5        | 65.3        | 66.3        | 33.5          | 48.7        | 51.7        | 52.5        | 50.9          | 55.3        | 60.6        | 60.1        |
| FS-DETR                           | <i>ICCV23</i>    | 44.6          | 50.8        | 52.7        | 56.4        | 37.3          | 43.4        | 46.4        | 48.9        | 43.8          | 50.6        | 52.1        | 56.9        |
| DiGeo-SMILe                       | <i>ECCV24</i>    | 36.1          | -           | 56.6        | 62.3        | 26.5          | -           | 44.3        | -           | 33.1          | -           | 51.9        | 56.4        |
| LLaFS <sup>†</sup>                | <i>CVPR24</i>    | 39.1          | 51.9        | 56.2        | 60.9        | 33.1          | 45.7        | 45.3        | 59.1        | 39.9          | 50.2        | 55.3        | 61.3        |
| FM-FSOD <sup>†</sup>              | <i>CVPR24</i>    | 41.6          | -           | 55.8        | 61.2        | 34.7          | -           | 47.6        | 58.7        | 39.5          | -           | 57.8        | 62.6        |
| LLMdet <sup>†</sup>               | <i>CVPR25</i>    | 39.9          | 51.7        | 56.1        | 60.8        | 33.8          | 45.6        | 45.4        | 59.0        | 39.7          | 49.9        | 55.1        | 61.0        |
| AgentDet-Llama3.1-8b <sup>†</sup> | (Ours)           | 49.2          | 59.7        | 64.2        | 64.3        | 38.2          | <b>50.9</b> | 51.9        | 53.4        | <b>53.2</b>   | 53.9        | 58.1        | 60.9        |
| AgentDet-Qwen2.5-7b <sup>†</sup>  | (Ours)           | <u>55.3</u>   | <u>63.2</u> | <u>68.3</u> | <u>69.3</u> | <u>39.7</u>   | 50.1        | <u>53.8</u> | <u>59.3</u> | 49.8          | <u>55.5</u> | <u>60.7</u> | <u>63.8</u> |
| AgentDet-Qwen3-8b <sup>†</sup>    | (Ours)           | <b>56.5</b>   | <b>64.2</b> | <b>68.5</b> | <b>69.6</b> | <b>39.8</b>   | 50.2        | <b>54.1</b> | <b>59.5</b> | <u>49.9</u>   | <b>55.7</b> | <b>60.9</b> | <b>63.9</b> |

Table 1. Few-shot object detection performance across the 3 splits on the PASCAL VOC benchmark, with 0-shot columns removed. The best and second-best results are bolded and underlined, respectively. The '†' indicates that the model incorporates VLMs or LLMs.

| Method                            |               | Shot       |             |             |             |             |             |
|-----------------------------------|---------------|------------|-------------|-------------|-------------|-------------|-------------|
|                                   |               | 0          | 1           | 3           | 5           | 10          | 30          |
| Meta-RCNN                         | <i>ICCV19</i> | -          | 1.0         | 2.8         | 4.0         | 6.5         | 11.1        |
| MPSR                              | <i>ECCV20</i> | -          | 5.1         | 7.4         | 8.7         | 9.8         | 14.1        |
| FSDetView                         | <i>ICML20</i> | -          | 4.5         | 7.2         | 10.7        | 12.5        | 14.7        |
| TFA                               | <i>CVPR21</i> | -          | 4.4         | 5.4         | 6.0         | 7.7         | 10.0        |
| Retentive R-CNN                   | <i>CVPR21</i> | -          | -           | -           | -           | 10.5        | 13.8        |
| CME                               | <i>CVPR21</i> | -          | -           | -           | -           | 15.1        | 16.9        |
| DCNet                             | <i>ICCV21</i> | -          | -           | -           | -           | 12.8        | 18.6        |
| QA-FewDet                         | <i>CVPR22</i> | -          | 4.9         | 8.4         | 9.7         | 11.6        | 16.5        |
| FCT                               | <i>AAAI22</i> | -          | 5.6         | 11.1        | 14.0        | 17.1        | 21.4        |
| Meta Faster R-CNN                 | <i>AAAI23</i> | -          | 5.1         | 9.8         | 10.8        | 12.7        | 16.1        |
| D&R                               | <i>ICCV23</i> | -          | 8.3         | 14.3        | 16.4        | 18.7        | 21.8        |
| FS-DETR                           | <i>ICCV23</i> | -          | 7.0         | 10.0        | 10.9        | 11.3        | -           |
| Norm-VAE                          | <i>CVPR23</i> | -          | 9.5         | 14.3        | 15.9        | 18.7        | 22.5        |
| PDC+SMILe                         | <i>ECCV24</i> | -          | -           | -           | -           | 25.8        | 31.0        |
| FM-FSOD <sup>†</sup>              | <i>CVPR24</i> | -          | 5.7         | 15.7        | 21.9        | 27.7        | 37.0        |
| LLMdet <sup>†</sup>               | <i>CVPR25</i> | 7.2        | 8.8         | 15.9        | 22.3        | 27.8        | <b>37.8</b> |
| AgentDet-Llama3.1-8b <sup>†</sup> | (Ours)        | 8.8        | 9.3         | <u>15.9</u> | 22.2        | 29.5        | 31.7        |
| AgentDet-Qwen2.5-7b <sup>†</sup>  | (Ours)        | 9.2        | <u>10.0</u> | 15.8        | <u>23.4</u> | <u>31.3</u> | 36.8        |
| AgentDet-Qwen3-8b <sup>†</sup>    | (Ours)        | <b>9.4</b> | <b>10.8</b> | <b>16.1</b> | <b>23.9</b> | <b>31.6</b> | <u>37.5</u> |

| Method (VLMs)               |  | Novel Split |             |             |
|-----------------------------|--|-------------|-------------|-------------|
|                             |  | 1           | 2           | 3           |
| InternVL2.5-7b              |  | 0.8         | 0.7         | 0.7         |
| LLaVA1.5-7b                 |  | 0.0         | 0.0         | 0.0         |
| Llama3.2-11b                |  | 1.2         | 1.4         | 1.8         |
| phi4-multimodal             |  | 0.1         | 0.0         | 0.2         |
| Qwen2.5-VL-7b               |  | 26.2        | 23.6        | 23.6        |
| Qwen2.5-VL-7b-ft            |  | 27.4        | 22.7        | 25.0        |
| AgentDet-Llama3.1-8b (Ours) |  | 30.2        | 24.9        | 31.0        |
| AgentDet-Qwen2.5-7b (Ours)  |  | <u>33.8</u> | <u>28.8</u> | 31.9        |
| AgentDet-Qwen3-8b (Ours)    |  | <b>35.7</b> | <b>29.2</b> | <b>33.5</b> |

Table 3. Zero-shot performance on the PASCAL VOC benchmark. The best and second-best results are bolded and underlined, respectively.

and PASCAL VOC comprises 20 categories. Few-shot setups split these datasets into base and novel classes, assessing the model’s ability to learn from limited data. See the supplementary material for more experimental details.

## 4.2. Analysis

We evaluate our approach on the PASCAL VOC and COCO datasets and compare it against the state-of-the-art few-shot object detection methods. Table 1 presents the performance of our method on the few-shot object detection task in the PASCAL VOC dataset, while Table 3 reports the results for the zero-shot object detection task. Table 2 further provides the evaluation metrics on the COCO dataset. Notably, since the pre-training data of the Qwen2.5-VL model includes

Table 2. Experiments on COCO dataset. The best and second-best results are bolded and underlined, respectively. The '†' indicates that the model incorporates VLMs or LLMs.

## 4. Experiments

### 4.1. Training Details

COCO and PASCAL VOC are standard benchmarks for few-shot object detection. COCO comprises 80 categories

part of the COCO dataset and Qwen2.5-VL has already demonstrated outstanding performance on RefCOCO, we do not consider it as our baseline on COCO.

On the PASCAL VOC dataset (Table 1), our method achieves the highest mAP across multiple novel splits. Specifically, AgentDet-Qwen3-8b attains 69.6% under the 10-shot setting in Novel Split 1. On the COCO dataset (Table 2), our method also exhibits superior performance. Under the 30-shot setting, AgentDet-Qwen3-8b reaches 37.5%, improving upon FM-FSOD (CVPR’24) by +0.5% but significantly outperforming other baselines. Notably, our method shows an advantage in low-shot regimes. These results validate the robustness of our LLM-based approach for few-shot detection.

Compared to modern text-based LLMs with billions of parameters, the additional CLIP, DINOv2 and related modules contribute only a small fraction of the total parameters, with the majority residing in the (mostly frozen) LLM itself. The KB capacity is explicitly bounded via our last-place elimination strategy, keeping memory usage modest. Thus, while the pipeline is structurally richer than a standard detector, it remains well-organized and parameter-efficient, and the added structure is necessary to fully exploit LLM capabilities in the low-shot regimes rather than as over-engineering.

### 4.3. Ablation

To evaluate the contribution of different components in our method, we conduct an ablation study on the PASCAL VOC dataset under the 10-shot setting. The results are summarized in Table 4. We evaluate the accuracy of the pre-processing stage on COCO and PASCAL VOC datasets. Allowing for a one-class margin of error, the accuracy of Agent-Scout reaches 89.67% on COCO and 93.97% on VOC as shown in Fig. 4.

**Effectiveness of Model Modules.** Table 4 presents the impact of key components in AgentDet-Qwen2.5-7b. Removing Q-former fine-tuning results in a significant 8.2% drop in mAP, while disabling knowledge base integration leads to a 19.1% decrease. If LLM fine-tuning is removed, the performance further drops drastically to 0%, highlighting the necessity of adapting the LLM to visual tasks. These findings confirm that each module in our framework plays a crucial role in enhancing few-shot detection performance.

**Effectiveness of Model Components.** Table 5 presents an ablation study evaluating the contribution of different components in AgentDet. Removing multi-scale segmentation reduces the accuracy at the query stage and training stage, highlighting its importance in multi-scale feature extraction. The removal of the minimum enclosing rectangle results in a significant decline to 49.7%. The filtering mechanism contributes to an improvement, as its removal lowers the accuracy to 61.3%.

| Method    | LLM   | QF-ft | KB | LLM-ft | Agent-Scout | mAP         |
|-----------|-------|-------|----|--------|-------------|-------------|
| TFA w/cos | -     | -     | -  | -      | -           | 36.8        |
| AgentDet  | Qwen  | ✓     | ✓  | ✓      | ✓           | <b>65.1</b> |
| AgentDet  | Llama | ✓     | ✓  | ✓      | ✓           | 60.4        |
|           |       | ✗     | ✓  | ✓      | ✓           | 52.2        |
|           |       | ✓     | ✗  | ✓      | ✓           | 41.3        |
|           |       | ✓     | ✓  | ✗      | ✓           | 0.0         |
|           |       | ✓     | ✓  | ✓      | ✗           | 30.2        |

Table 4. Ablation study of different modules in AgentDet.

| Method                                         | mAP         |
|------------------------------------------------|-------------|
| AgentDet                                       | <b>65.1</b> |
| AgentDet query w/o Multi-Scale Segmentation    | 56.9        |
| AgentDet training w/o Multi-Scale Segmentation | 55.4        |
| AgentDet w/o Minimum Enclosing Rectangle       | 49.7        |
| AgentDet w/o Filter                            | 61.3        |

Table 5. Ablation study on different components.

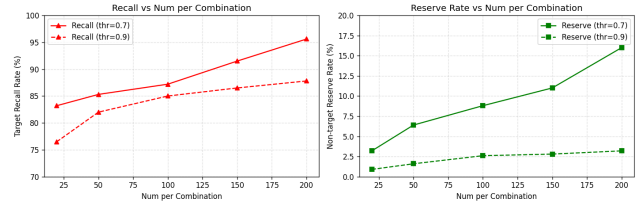


Figure 4. Effect of crop density and threshold on category detection. Red curves show the average target recall rate; green curves show the non-target filter rate. Solid and dashed lines correspond to thresholds 0.7 and 0.9, respectively.

## 5. Conclusion

In this paper, we present AgentDet, a unified framework that leverages large language models and structured visual knowledge for zero- and few-shot object detection. Through the integration of four agents, AgentDet achieves strong performance on standard benchmarks. Extensive experiments and ablations validate the effectiveness of each component. Our work highlights the potential of LLMs in open-vocabulary detection and provides a promising direction for future research in vision-language understanding.

## Acknowledgements

This work was supported by the Science and Technology Program of Qingdao Municipality under grant 25-1-1-gjgg-32-gx.



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